



NEET CHALLENGER 2025

Physics - 45		Chemistry - 45	Biology - 45
Mark 720	Group PCB	PCB CPT EXAM - 09	
		Date: 17/08/2025	Time: 3:00 Hours

Question Booklet Ver	Question Booklet Sr. No
P (Write this number on your Answer Sheet)	4325

This is to certify that the answers of PCB CPT EXAM - 09 have been correctly written and verified.

Invigilator's Signature

- This question paper contains Multiple Choice Questions (MCQs) of Physics [Questions: 45] and 1/4th MARKS. Total score will be 45.
- Chemistry [Questions: 45] and 1/4 MARKS (ONE). Total score will be 45.
- Biology [Questions: 45] and 1/4 MARKS (ONE). Total score will be 45.
- The question paper is to be read separately at the start of the examination.
- Choice and sequence of questions is at the discretion of the candidate.
- Candidate should carefully read the instructions printed on the Question Booklet and Answer Sheet and make the correct entries on the Answer Sheet. Candidate should be taken to fill QUESTION BOOKLET VERSION, SERIAL No. and NEET-2025 Roll no. on the Answer Sheet and Question Booklet.
- Determine the one correct answer from out of the four available options given for each question.
- Fill the appropriate circle completely like this ● for answering a particular question. Mark with **Black ink ball point pen only**.
- Use of whitener or any other material to erase/hide the circle once filled is not permitted.**
- Avoid overwriting and/or striking of answers once marked.
- Rough work should be done only on the blank space provided on the Question Booklet. Rough work should not be done on the Answer Sheet.
- Immediately after the prescribed examination time is over Answer sheet is to be returned to the invigilator. Confirm that the candidate and invigilator have signed on question booklet and Answer Sheet.
- The candidate is allowed to leave the examination hall till the end of examination.

Test Syllabus (NEET REPT 2025-26 26 May Batch)

Physics
Chemistry
Biology

Circular Motion, Work, Power & Energy
Chemical Bonding, GOC-Electronic Effect
Photosynthesis in Higher Plants, Body fluids & Circulation

PHYSICS

01. A ball is released from the top of a tower. The ratio of work done by force of gravity in first, second and third second of the motion of the ball is:

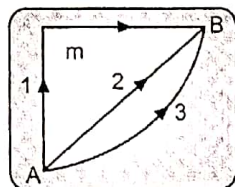
(1) $1 : 2 : 3$

(2) $1 : 4 : 9$

(3) $1 : 3 : 5$

(4) $1 : 5 : 3$

02. If W_1 , W_2 and W_3 represent the work done in moving a particle from A to B along three different paths 1, 2 and 3 respectively (as shown) in the gravitational field of a point mass m , find the correct relation between W_1 , W_2 and W_3 :



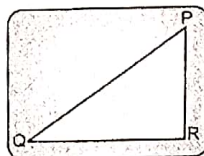
(1) $W_1 > W_2 > W_3$

(2) $W_1 = W_2 = W_3$

(3) $W_1 < W_2 < W_3$

(4) $W_2 > W_1 > W_3$

03. For the path PQR in a conservative force field (fig), the amounts of work done in carrying a body from P to Q and from Q to R are 5 J and 2 J respectively. The work done in carrying the body from R to P will be:



(1) -7 J

(2) -3 J

(3) -5 J

(4) zero

04. A particle moves in a circle of radius 5 cm with constant speed and time period $0.2\pi \text{ s}$. The acceleration of the particle is:

(1) 25 m/s^2

(2) 36 m/s^2

(3) 5 m/s^2

(4) 15 m/s^2

05. The angular speed of flywheel making 360 rev/min is:

(1) $12 \pi \text{ rad/s}$

(2) $6 \pi \text{ rad/s}$

(3) $3 \pi \text{ rad/s}$

(4) $2 \pi \text{ rad/s}$

06. An electric fan has blades of length 30 cm as measured from the axis of rotation. If the fan is rotating at 1200 rpm, the acceleration of a point on the tip of the blade is about:

(1) 1600 cm^{-2}

(2) 4740 ms^{-2}

(3) 2370 ms^{-2}

(4) 5055 ms^{-2}

07. Work done by force of friction (i) can be zero (ii) can be positive (iii) can be negative (iv) is always negative.

The correct statement are:

(1) i and ii

(2) i and iv

(3) only iv

(4) i, ii and iii

08. Which of the following force is non-conservative?

(1) Spring force

(2) Coulomb force

(3) Gravitational force

(4) Viscous force

09. The angular velocity of second hand of a clock is:

(1) $\left(\frac{\pi}{6}\right) \text{ rad s}^{-1}$

(2) $\left(\frac{\pi}{60}\right) \text{ rad s}^{-1}$

(3) $\left(\frac{\pi}{30}\right) \text{ rad s}^{-1}$

(4) $\left(\frac{\pi}{15}\right) \text{ rad s}^{-1}$

10. The acceleration of a body in a non-uniform circular motion is 5 m/s^2 . Which one of the following is correct?

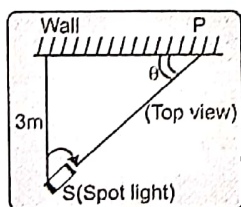
(1) The radial acceleration and the tangential accelerations are 3 ms^{-2} and 4 ms^{-2} respectively.

(2) The radial and the tangential accelerations are 2 ms^{-2} and 3 ms^{-2} respectively.

(3) The radial and the tangential accelerations are both 5 ms^{-2} .

(4) The radial and the tangential acceleration are 5 ms^{-2} and 3 ms^{-2} respectively.

11. A particle is moved from $(0,0)$ to $(2m, 2m)$ with force $(\vec{F} = (x\hat{j}) \text{ N})$ along path $y = x$. Find the work done
 (1) 4 J (2) 2 J
 (3) $\frac{8}{3}$ J (4) None.
12. A spring of spring constant 5000 N/m is stretched initially by 5 cm from the unstretched position. The work required to stretch it further by another 5 cm is:
 (1) 6.25 N-m
 (2) 12.50 N-m
 (3) 18.75 N-m
 (4) 25.00 N-m
13. The work done by a force, $\vec{F} = (-6x^3)\hat{i}$ N in displacing a particle from $x = 4$ to $x = -2$ m is:
 (1) 360 J (2) 240 J
 (3) -240 J (4) -360 J
14. The position vector of a particle $\vec{R}$ as a function of time is given by, $\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$
 Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ demite unit vectors along x and y -directions, respectively. Which one of the following statements is wrong for the motion of particles?
 (1) Path of the particle is a circle of radius 4 meter.
 (2) Acceleration vectors is along $-\vec{R}$.
 (3) Magnitude of acceleration vector is $\frac{v^2}{R}$ where v is the velocity of particle.
 (4) Magnitude of the velocity of particle is 8 meter/second.
15. A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s . The spot of light P moves along the wall at a distance 3m. What is the velocity of the spot P when $\theta = 45^\circ$?



- (1) 0.6m/s (2) 0.5m/s
 (3) 0.4m/s (4) 0.3m/s

16. The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as :

- (1) $2a \sin \omega t$ (2) $2a \sin \frac{\omega t}{2}$
 (3) $2a \cos \omega t$ (4) $2a \cos \frac{\omega t}{2}$

17. An elastic string of unstretched length L and force constant k is stretched by a small length x . It is further stretch by small length y . The work done in the second stretching is:

- (1) $\left(\frac{1}{2}\right)ky^2$ (2) $\left(\frac{1}{2}\right)k(x^2 + y^2)$
 (3) $\left(\frac{1}{2}\right)k(x+y)^2$ (4) $\left(\frac{1}{2}\right)ky(2x+y)$

18. When a rubber-band is stretched by a distance x , it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:

- (1) $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$
 (2) $aL^2 + bL^3$
 (3) $\frac{1}{2}(aL^2 + bL^3)$

(4) $\frac{aL^2}{2} + \frac{bL^3}{3}$

19. One end of a light spring of spring constant k is fixed to a wall and the other end is tied to a block placed on a smooth horizontal surface. In a displacement, the work done by the spring is $+\frac{1}{2}kx^2$.

The possible cases are:

- (1) The spring was initially compressed by a distance x and was finally in its natural length.
 (2) It was initially elongated a distance x and finally was in its compressed by distance x .
 (3) It was initially in its natural length and finally in a compressed position.
 (4) It was initially in its natural length and finally in a stretched position.

Date :- 17/08/2025

20. Work done by spring force of a certain spring when stretched through a distance 'S' is 10 joule. The amount of work (in joule) that a spring force will do to stretch it through an additional distance 'S' is:

(1) 30
(2) 40
(3) 10
(4) 20

21. A particle covers equal distance around a circular path, in equal intervals of time. Which of the following quantities connected with the motion of the particle remains constant with time?

(1) Displacement
(2) Velocity
(3) Speed
(4) Acceleration

22. A car is travelling at 50 m/s on a circular road of radius 1000 m. It is increasing its speed at the rate of 4 m/s^2 . The acceleration of the car is:

(1) 2.5 m/s^2
(2) 6.5 m/s^2
(3) 4.7 m/s^2
(4) 5.7 m/s^2

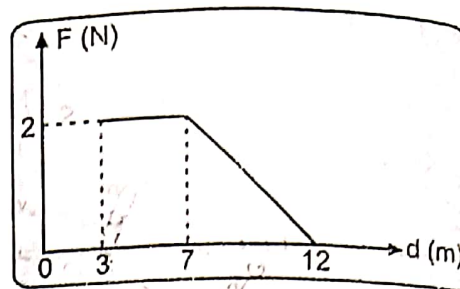
23. A particle is moving along a circular path with a uniform speed. When the particle completes half of the circular path, find the angle through which the angular velocity of particle will change:

(1) 0°
(2) 30°
(3) 45°
(4) 180°

24. A body of mass m is moving on a circular path of radius r with a constant speed v . The work done by the centripetal force in moving the body over half the circumference of the circle is:

(1) $2mv^2$
(2) $m v^2 r$
(3) $mv^2 / 2r$
(4) zero

25. Force F on a particle moving in a straight line varies with distance d as shown in the figure. The work done on the particle during its displacement of 12 m is:



(1) 13J
(2) 18J
(3) 21J
(4) 26J

26. Two cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that they make complete circles in the same time t . The ratio of their centripetal acceleration is:

(1) $m_1 r_1 : m_2 r_2$
(2) $m_1 : m_2$
(3) $r_1 : r_2$
(4) 1 : 1

27. If the length of second's hand in a stop clock is 3 cm. The angular velocity and linear velocity of the tip is

(1) 0.2047 rad/s , 0.0314 m/s
(2) 0.2547 rad/s , 0.314 m/s
(3) 0.1472 rad/s , 0.06314 m/s
(4) 0.1047 rad/s , 0.00314 m/s

28. A particle moves with constant speed v along a circular path of radius r and completes the circle in time T . The acceleration of the particle is

(1) $2\pi v / T$
(2) $2\pi r / T$
(3) $2\pi r^2 / T$
(4) $2\pi v^2 / T$

29. The work done by a force $\vec{F} = 3\hat{i} - 4\hat{j} + 5\hat{k}$ displaces the body from a point (3,4,6) to a point (7,2,5) is

(1) 15 units
(2) 25 units
(3) 20 units
(4) 10 units

Date :-17/08/2025

30. No work is done by a force on an object if.
- (i) the force is always perpendicular to its velocity
 - (ii) the force is always perpendicular to its acceleration
 - (iii) the object is stationary but the point of application of the force moves on the object.
 - (iv) the object moves in such a way that the point of application of the force remains fixed.

The correct statement are

- (1) i, ii
 - (2) i, iii, iv
 - (3) iii, iv
 - (4) i, ii, iii, iv
31. A stone tied to the end of a string 80 cm long is whirled in a horizontal circle with a constant speed. If the stone makes 14 revolutions in 25 s, what is the magnitude of the acceleration of stone ?

- (1) 0.99 cm/s²
- (2) 9.9 cm/s²
- (3) 0.99 m/s²
- (4) 9.9 m/s²

32. An aircraft executes a horizontal loop of radius 1.00 km with a steady speed of 900 km/h. Compare its centripetal acceleration with the acceleration due to gravity.

- (a) 0.638
- (2) 6.38
- (3) 63.8
- (4) 36.8

33. A force $\vec{F} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ acts on a particle and produces a displacement of $\vec{s} = 2\hat{i} - 3\hat{j} + x\hat{k}$. If the done is zero, the value of x is:

- (1) -2
- (2) 1/2
- (3) 6
- (4) 2

34. Two masses of 1 g and 9 g are moving with equal kinetic energies. The ratio of the magnitudes of their respective linear momenta is:

- (1) 1 : 9
- (2) 9 : 1
- (3) 1 : 3
- (4) 3 : 1

35. A car is moving along a circular path of radius 500 m with a speed of 30 m/s. If at some instant, its speed increases at the rate of 2 m/s², then at that instant the magnitude of resultant acceleration will be :

- (1) 4.7 m/s²
- (2) 3.8 m/s²
- (3) 3 m/s²
- (4) 2.7 m/s²

36. A car of mass m moves in a horizontal circular path of radius r metre. At an instant its speed is V m/s and is increasing at a rate of a m/sec². Then, the acceleration of the car is :

- (1) $\frac{V^2}{r}$
- (2) a
- (3) $\sqrt{a^2 + \left(\frac{V^2}{r}\right)^2}$
- (4) $\sqrt{a + \frac{V^2}{r}}$

37. If the momentum of a body is increased by 50% then the percentage increase in its kinetic energy is:

- (1) 50 %
- (2) 100 %
- (3) 125 %
- (4) 200 %

38. A stone is thrown at angle 45° to the horizontal with kinetic energy K . The kinetic energy at the highest point is:

- (1) $\frac{K}{2}$
- (2) $\frac{\sqrt{K}}{2}$
- (3) K
- (4) zero

39. If the kinetic energy of a particle is increased by 300%, the momentum of the particle will increase by:

- (1) 20%
- (2) 200%
- (3) 100%
- (4) 50%

Date :-17/08/2025



40. When a man increases his speed by 2 ms^{-1} , he finds that his kinetic energy is doubled the original speed of the man is:

- (1) $2(\sqrt{2} - 1) \text{ ms}^{-1}$
- (2) $2(\sqrt{2} + 1) \text{ ms}^{-1}$
- (3) 4.5 ms^{-1}
- (4) none of these

41. If a_r and a_t represent radial and tangential accelerations, the motion of the particle will be uniformly circular if

- (1) $a_r = 0$ and $a_t = 0$
- (2) $a_r = 0$ and $a_t \neq 0$
- (3) $a_r \neq 0$ and $a_t = 0$
- (4) $a_r \neq 0$ and $a_t \neq 0$

42. For a particle in uniform circular motion, the acceleration $\vec{a}$ at a point $P(R, \theta)$ on the circle of radius R is (Here θ is measured from x-axis)

- (1) $\frac{v^2}{R} \hat{i} + \frac{v^2}{R} \hat{j}$
- (2) $-\frac{v^2}{R} \cos \theta \hat{i} + \frac{v^2}{R} \sin \theta \hat{j}$
- (3) $-\frac{v^2}{R} \sin \theta \hat{i} + \frac{v^2}{R} \cos \theta \hat{j}$
- (4) $-\frac{v^2}{R} \cos \theta \hat{i} - \frac{v^2}{R} \sin \theta \hat{j}$

43. If a cycle wheel of radius 4m completes one revolution in two seconds. Then acceleration of a point on the cycle wheel will be

- (1) $\pi^2 \text{ m/s}^2$
- (2) $2\pi^2 \text{ m/s}^2$
- (3) $4\pi^2 \text{ m/s}^2$
- (4) $8\pi^2 \text{ m/s}^2$

44. If the momentum of a body is increased n times, its kinetic energy increases:

- (1) n times
- (2) $2n$ times
- (3) $\sqrt{n}$ times
- (4) n^2 times

45. If the kinetic energy of a body increases by 0.1%, the percent increase of its momentum will be:

- (1) 0.05 %
- (2) 0.1 %
- (3) 1.0 %
- (4) 10 %

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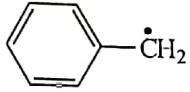
CHEMISTRY

46. The stability of caranions in the following:

- (A) $\text{RC} \equiv \text{C}^-$ (B) $\text{R}_2\text{C} = \text{CH}^-$
 (C)  (D) $\text{R}_3\text{C}-\text{CH}_2^-$

is in the order of:

- (1) (D) > (B) > (C) > (A)
 (2) (A) > (C) > (B) > (D)
 (3) (A) > (B) > (C) > (D)
 (4) (B) > (C) > (D) > (A)

47. The radical,  is aromatic because it has:

- (1) 6p-orbitals and 7 unpaired electrons
 (2) 6p-orbitals and 6 unpaired electrons
 (3) 7p-orbitals and 6 unpaired electrons
 (4) 7p-orbitals and 6 unpaired electrons

48. The dipole moment of o, p and m-dichlorobenzene will be in the order

- (1) $o > p > m$ (2) $p > o > m$
 (3) $m > o > p$ (4) $o > m > p$

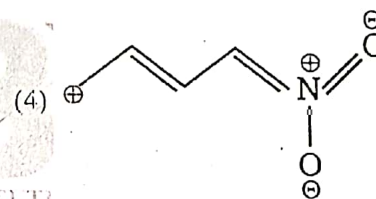
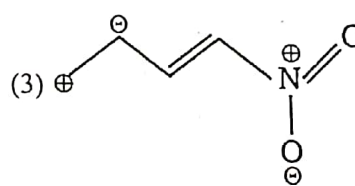
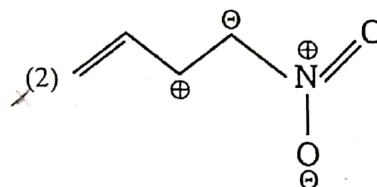
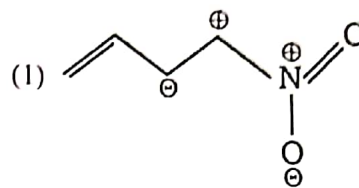
49. What is not true about resonance?

- (1) The resonating structures are hypothetical
 (2) The unpaired electrons in various resonating structures are same
 (3) Hybrid structure is least stable
 (4) Hybrid structure is least energetic

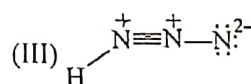
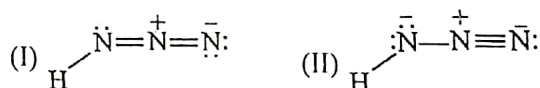
50. Which one is the correct order of acidity?

- (1) $\text{CH} \equiv \text{CH} > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH} > \text{CH}_3 - \text{CH} = \text{CH} > \text{CH}_3 - \text{CH}_3$
 (2) $\text{CH} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_3 - \text{CH} = \text{CH} > \text{CH}_3 - \text{CH}_3$
 (3) $\text{CH}_3 = \text{CH}_3 > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_3 - \text{CH} = \text{CH} > \text{CH}_3 - \text{CH}_3$
 (4) $\text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{CH} = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_3 - \text{CH} = \text{CH} > \text{CH}_3 - \text{CH}_3$

51. Among the following, the least stable resonance structure is:



52. Which of the following structures of hydrazoic acid molecule is not expected to be canonical form?

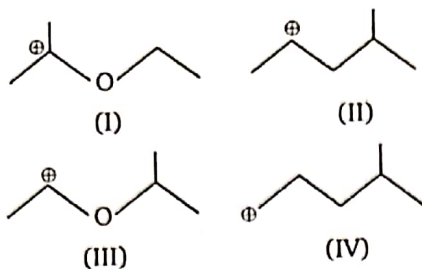


- (1) I only (2) II only
 (3) Both I and II (4) III only

53. Arrange the following compounds in order of increasing dipole moment:

- (I) Toluene (II) m-Dichlorobenzene
 (III) o-Dichlorobenzene (IV) p-Dichlorobenzene
 (1) I < IV < II < III
 (2) IV < I < II < III
 (3) IV < I < III < II
 (4) IV < II < I < III

54. The correct stability order for the following species is:



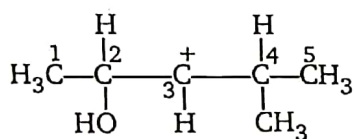
- (1) (II) > (IV) > (I) > (III)
 (2) (I) > (II) > (III) > (IV)
 (3) (II) > (I) > (IV) > (III)
 (4) (I) > (III) > (II) > (IV)
55. Hyperconjugation involves overlap of the following orbitals:

- (1) $\sigma - \sigma$ (2) $\sigma - p$
 (3) $p - p$ (4) $\pi - \pi$

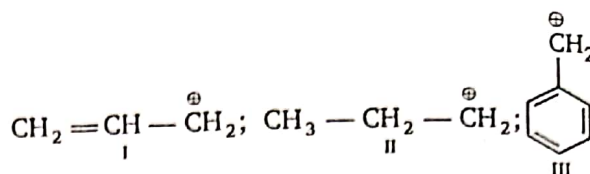
56. The table shown below gives the bond dissociation energies (E_{diss}) for single covalent bonds of carbon (C) atoms with element A, B, C and D. Which element has the smallest atoms?

Bond	E_{diss} (kJ mol^{-1})
C-A	240
C-B	328
C-C	276
C-D	485

- (1) C (2) D
 (3) A (4) B
57. In which of the following the angle between the two covalent bonds is largest?
- (1) H_2O (2) NH_3
 (3) CO_2 (4) CH_4
58. In the following carbocation, H/CH_3 that is most likely to migrate to the positive charged carbon is:



59. The order of stability of the following carbocations:



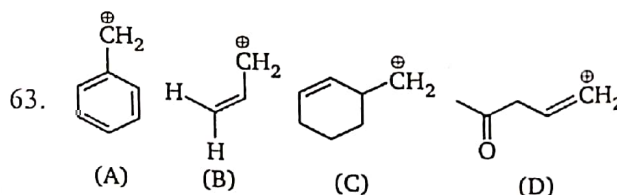
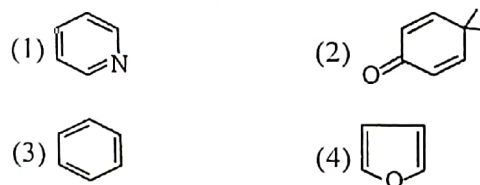
60. The shape of the compounds like ClF_3 , BrF_5 , XeF_4 and SF_4 , respectively, are

- (1) see-saw, square planar, square pyramidal, T-shaped
 (2) T-shaped, square planar, bent, see-saw
 (3) T-shaped, square pyramidal, square planar, see-saw
 (4) square planar, see-saw, T-shaped, square pyramidal

61. In BF_3 , the B-F bond length is 1.30 Å. when BF_3 is allowed to be treated with Me_3N , it forms an adduct, $\text{Me}_3\text{N} \rightarrow \text{BF}_3$, the bond length of B-F in the adduct is

- (1) Greater than 1.30 Å
 (2) Smaller than 1.30 Å
 (3) Equal to 1.30 Å
 (4) None of these

62. Which of the following molecules is least resonance stabilised?



Among the given species the Resonance stabilised carbocations are:

- (1) (C) and (D) only (2) (A), (B) and (D) only
 (3) (A) and (B) only (4) (A), (B) and (C) only

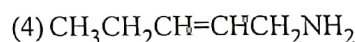
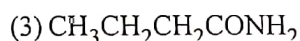
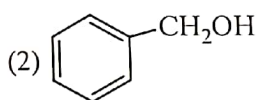
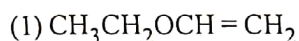
64. According to molecular orbital theory which of the following is correct?

- (1) C_2 molecule is diamagnetic
- (2) C_2^- ion is paramagnetic
- (3) Bond order of C_2 molecule is 2
- (4) All of the above

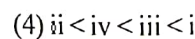
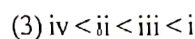
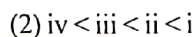
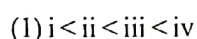
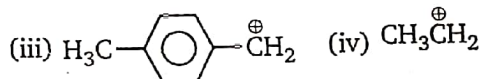
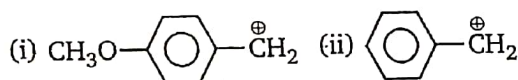
65. In a homonuclear molecule which of the following set of orbitals are degenerate?

- (1) σ_{1s} and σ_{2s}
- (2) π_{2p_x} and π_{2p_y}
- (3) π_{2p_x} and σ_{2p_z}
- (4) σ_{2p_z} and $\sigma^*_{2p_z}$

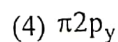
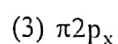
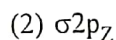
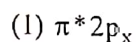
66. Which of the following compounds does not exhibit resonance?



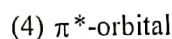
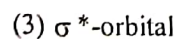
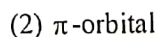
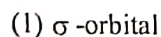
67. Relative stabilities of the following carbocations will be in the order:



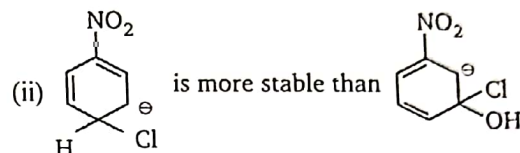
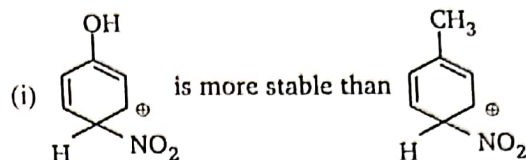
68. Two nodal planes are present in



69. During change of $NO^+ \rightarrow NO$, the electron is added to



70. Consider the following statements:



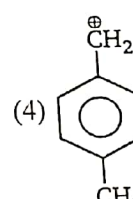
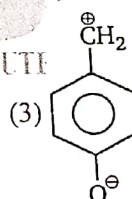
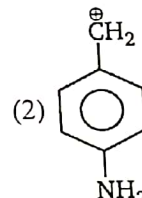
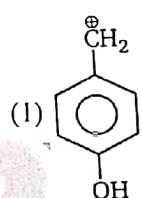
(1) i and ii both are correct

(2) i and reverse of ii are correct

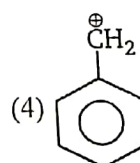
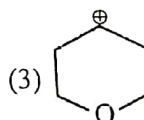
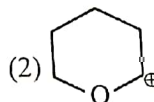
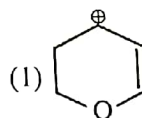
(3) ii and reverse of i are correct

(4) i and ii both are incorrect

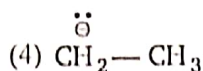
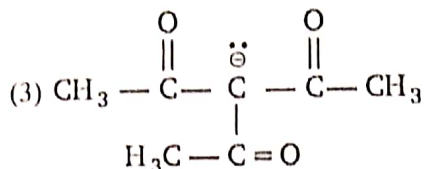
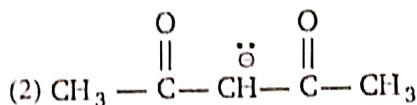
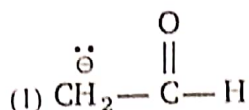
71. Which of the following carbocation is most stable?



72. Which of the following is most stable carbocation?



73. Most stable carbanion is:



74. Consider the statements

I. Bond length in N_2^+ is $0.02 \text{ \AA}$ greater than in N_2 .

II. Bond length in NO^+ is $0.09 \text{ \AA}$ less than in NO .

III. O_2^{2-} has a shorter bond length than O_2 .

Then which of the following is correct?

(1) I only

(2) II and III

(3) I and II

(4) I, II and III

75. The energy of σ_{2s} -orbital is greater than σ^*_{1s} orbital because

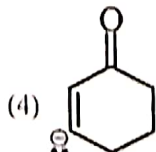
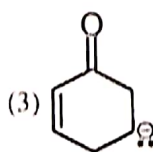
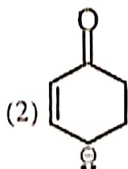
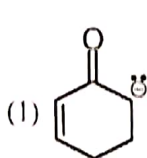
(1) σ_{2s} orbital is bigger than σ_{1s} orbital

(2) σ_{2s} orbital is a bonding orbital where, σ^*_{1s} is an antibonding orbital

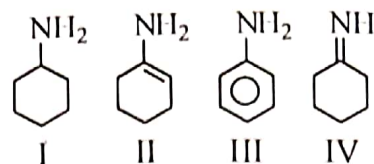
(3) σ_{2s} orbital has a greater value of n than σ^*_{1s} orbital

(4) None of the above

76. Most stable carbanion is:



77. Correct order of bond length between C-N



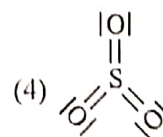
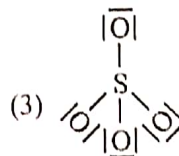
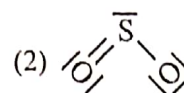
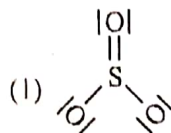
(1) I > II > III > IV

(2) IV > III > II > I

(3) III > IV > I > II

(4) II > III > IV > I

78. Which of the following structures is the most preferred and hence of lowest energy for SO_3 ?



79. Four diatomic species are listed below. Identify the correct order in which the bond order is increasing in them

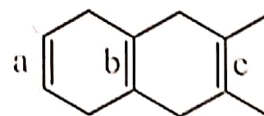
(1) $\text{NO} < \text{O}_2 < \text{C}_2^+ < \text{He}_2^+$

(2) $\text{O}_2 < \text{NO} < \text{C}_2^+ < \text{He}_2^+$

(3) $\text{C}_2^+ < \text{He}_2^+ < \text{O}_2 < \text{NO}$

(4) $\text{He}_2^+ < \text{O}_2 < \text{NO} < \text{C}_2^+$

80. Compare the bond length between C=C



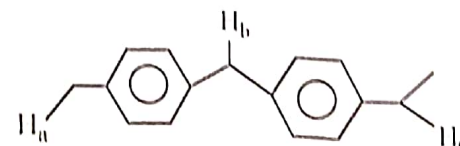
(1) $a > b > c$

(2) $c > a > b$

(3) $a > c > b$

(4) $c > b > a$

81.



Compare bond strength

(1) $a > b > c$

(2) $a > c > b$

(3) $b > a > c$

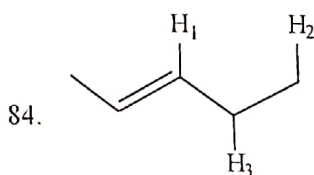
(4) $c > b > a$

82. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is

- (1) $O_2^{2-} > O_2^- > O_2 > O_2^+$
- (2) $O_2 > O_2^+ > O_2^{2-} > O_2^-$
- (3) $O_2^- > O_2^{2-} > O_2^+ > O_2$
- (4) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

83. Some of the properties of the two species, NO_3^- and H_3O^+ are described below. Which one of them is correct?

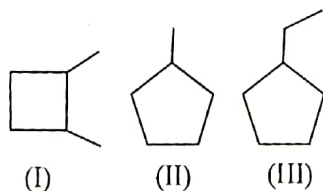
- (1) Dissimilar in hybridization for the central atom with different structures
- (2) Isostructural with same hybridization for the central atom
- (3) Isostructural with different hybridization for the central atom
- (4) Similar in hybridization for the central atom with different structures



Compare bond energy

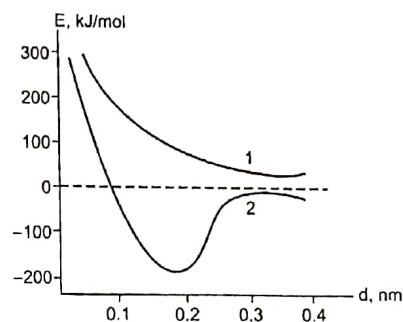
- (1) $H_1 > H_2 > H_3$
- (2) $H_3 > H_2 > H_1$
- (3) $H_2 > H_1 > H_3$
- (4) $H_1 > H_3 > H_2$

85. Correct order of heat of combustion-



- (1) $III > II > I$
- (2) $II > I > III$
- (3) $III > I > II$
- (4) $I > II > III$

86. Consider the given figure showing the formation of H_2^+ ion depending on internuclear distance versus potential energy of the system.

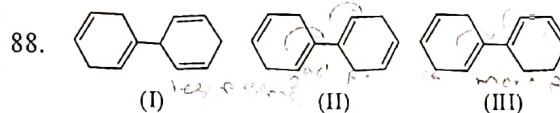


which is correct statement :

- (1) Curve-1 represents the most stable state of the system for H_2^+ ion
- (2) Curve-2 represents the most stable state of the system for H_2^+ ion
- (3) Curve-1 indicates that the molecular hydrogen ion is formed.
- (4) Curve-2 represents the energy level of the antibonding region.

87. The shape of methyl radical ($\dot{C}H_3$) is

- (1) Linear
- (2) Pyramidal
- (3) Planar
- (4) Spherical



- (1) $I > II > III$
- (2) $III > II > I$
- (3) $I > III > II$
- (4) $III > I > II$

89. Which of the following conditions is not correct for resonating structures ?

- (1) The contributing structures must have the same number of unpaired electrons
- (2) The contributing structures should have almost similar energies.
- (3) The contributing structures should be so written that unlike charges reside on atoms that are far apart
- (4) The positive charge should be present on the electropositive element and the negative charge on the electronegative element.

90. The number of three centre two electron bonds in a molecule of diborane is

- (1) 0
- (2) 2
- (3) 4
- (4) 6

Date :-17/08/2025

BIOLOGY

91. Photosystem is composed of
 (1) Light harvesting complex
 (2) Reaction centre
 (3) Accessory pigments
 (4) More than one option is correct
92. The oxygen liberated during the photosynthesis comes from
 (1) CO_2
 (2) Water
 (3) Photosynthetic enzymes
 (4) Carbohydrates
93. Read the following statements:
 (1) F_0 part of ATP synthase is associated with breakdown of proton gradient.
 (2) A H-carrier contributes in creation of proton gradient.
 (3) Movement of electrons in ETS is coupled to pumping of protons into the lumen.
 (4) Formation of $\text{NADPH} + \text{H}^+$ is related with the creation of proton gradient. How many of the above statements are correct?
 (1) Two (2) One
 (3) Four (4) Three
94. The volume of blood present in an adult human is
 (1) 10 litres (2) 5 litres
 (3) 2 litres (4) 1 litre.
95. The major proteins of blood plasma are
 (1) Albumin, globulin and fibrinogen
 (2) Collagen, elastin and fibrinogen
 (3) Albumin, glutelin and p -protein
 (4) Actin, myosin and tubulin.
96. Thromboplastin is produced by
 (1) Damaged tissues
 (2) Blood platelets
 (3) Both blood platelets and damaged tissues
 (4) Prothrombin.
97. The life span of the erythrocytes in mammalian blood is about
 (1) 120 days (2) 150 days
 (3) 190 days (4) 180 days
98. Which of the following chemicals can be used as anticoagulant?
 (1) Sodium chloride
 (2) Sodium citrate
 (3) Sodium nitrate
 (4) Ammonium chloride.
99. Which among the following is not a step in Calvin cycle?
 (1) Carboxylation
 (2) Reduction
 (3) Photophosphorylation
 (4) Regeneration
100. The enzyme RuBP carboxylase
 (1) Activity occurs in C_3 and C_4 plants
 (2) It present in inner thylakoid membrane
 (3) Is low temperature sensitive enzyme
 (4) Shows greater affinity for O_2 than for CO_2
101. Which of the following statements is not true regarding the C_4 plants?
 (1) They show Kranz anatomy
 (2) Decarboxylation process occurs in bundle sheath cells
 (3) Granal chloroplasts are present in bundle sheath cells
 (4) PEPcase enzyme activity occurs in mesophyll cells
102. The C_4 and C_3 plants differ from each other in
 (1) Type of pigments involved in photosynthesis
 (2) The primary acceptor of CO_2 during carbon fixation
 (3) Type of end products of photosynthesis
 (4) Number of NADPH that are consumed during the glucose synthesis process
103. Which of the following statement is correct regarding the C_4 plants?
 (1) The carbon fixation is catalysed by PEP carboxylase
 (2) Mesophyll cells lack the RuBisCO enzyme
 (3) Photorespiration is absent
 (4) More than one option is correct
104. Ribulose 1,5 biphosphate carboxylase enzyme catalyses the oxygenation reaction between
 (1) PGA and dihydroxyacetone phosphate
 (2) oxaloacetic acid and acetyl CoA
 (3) CO_2 and Ribulose 1,5 biphosphate
 (4) O_2 and Ribulose 1,5 biphosphate

Date :-17/08/2025



105. Which of the following is not a feature of C_3 plants?

- (1) They cannot tolerate higher temperatures and higher light intensities.
- (2) Photorespiration operates in them
- (3) They do not have greater productivity of biomass
- (4) They show C_3 cycle in bundle sheath cells

106. ATP synthesis in chloroplast and mitochondria is due to the proton gradient across the membrane. Select the correct statements w.r.t ATP formation in chloroplast

- (A) Proton accumulates in the lumen of thylakoid
- (B) Splitting of water occurs on the inner side of the membrane
- (C) Proton accumulates in the stroma of chloroplast
- (D) NADP reductase is located on the stroma side of the membrane

- (1) Only A and B are correct
- (2) Only B and C are correct
- (3) Only A, B and D are correct
- (4) All are correct

107. What is the advantage of light reactions producing ATP and NADPH on stromal side of thylakoid membrane?

- (1) Calvin cycle consumes ATP and NADPH from stroma
- (2) Light reaction occurs in stroma
- (3) Dark reaction occurs in grana need ATP + NADPH
- (4) CO_2 is produced in stroma

108. The first step in photosynthesis is

- (1) Excitation of chlorophyll by light
- (2) Ionisation of water
- (3) ATP synthesis
- (4) Production of assimilatory power

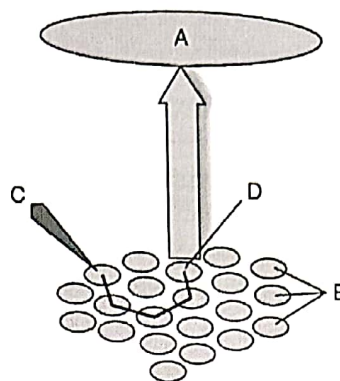
109. Select the incorrect pair:

- (1) 3-carbon compound – Phosphoglyceric acid
- (2) 2-carbon compound – Malic acid
- (3) 4-carbon compound – Oxaloacetic acid
- (4) 5-carbon compound – Ribulose 1, 5 Bisphosphate

110. For easy understanding, the Calvin cycle is divided into how many stages?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

111. Identify A to D in the given figure.



- (1) A–Pigment molecules, B–Primary acceptor, C–Photon, D–Reaction centre
- (2) A–Photon, B–Primary acceptor, C–Reaction centre, D–Pigment molecules
- (3) A–Primary acceptor, B–Pigment molecules, C–Photon, D–Reaction centre
- (4) A–Reaction centre, B–Primary acceptor, C–Pigment molecules, D–Photon

112. In C_3 plants, the first stable product of photosynthesis during dark reaction is

- (1) Phosphoglyceric acid
- (2) Phosphoglyceraldehyde
- (3) Malic acid
- (4) Oxaloacetic acid

113. How many turns of Calvin cycle forms one glucose molecule?

- (1) 2
- (2) 6
- (3) 4
- (4) 8

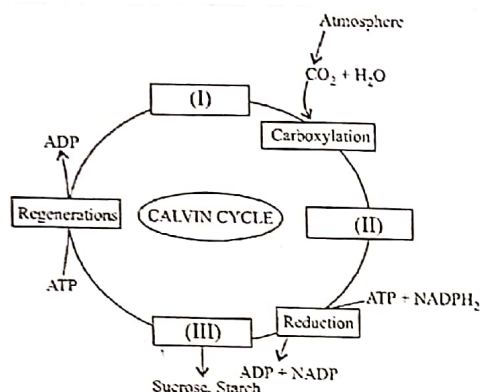
114. In C_4 plants, the mesophyll cells are connected with bundle sheath cells with the help of

- (1) Cytoplasmic connection
- (2) Special connecting tissues
- (3) Plasmodesmata
- (4) Connection is not essential

115. In C_4 plants, Calvin cycle occurs in

- (1) Stroma of bundle sheath chloroplast
- (2) Mesophyll chloroplast
- (3) Grana of bundle sheath chloroplast.
- (4) Does not occur as CO_2 is fixed mainly by PEP and no CO_2 is left for the Calvin cycle.

116. Which of the following plant shows oxaloacetic acid as the first stable product?
- Tomato
 - Rice
 - Sorghum
 - Bell pepper
117. The given figure of Calvin cycle shows the carbon assimilation in C_3 plants. Choose the correct labelling of the carbohydrate molecule (Marked as I, II and III) involved in the Calvin cycle.



- (I)-RuBP, (II)-Triose phosphate, (III)-PGA
 - (I)-PGA, (II)-RuBP, (III)-Triose phosphate
 - (I)-PGA, (II)-Triose phosphate, (III)-RuBP
 - (I)-RuBP, (II)-PGA, (III)-Triose phosphate
118. Which one of the following is wrong in relation to photorespiration?
- It is a characteristic of C_3 plants
 - It occurs in chloroplasts
 - It occurs in day time only
 - It is a characteristic of C_4 plants
119. Agranulocytes are
- Eosinophils
 - Neutrophils
 - Basophils
 - None of the above
120. Blood does not contain
- Calcium
 - Prothrombin
 - Fibrinogen
 - Elastin.
121. The metal essential for haemoglobin formation is
- Mg^{++}
 - Na^{++}
 - Fe^{++}
 - Cu^{++}

122. Cells which help in the transportation of O_2 are
- WBC
 - Leucocytes
 - RBC
 - Thrombocytes.
123. The respiratory pigment present in the RBC of vertebrate blood is called
- Haemoglobin
 - Haematin
 - Haematochrome
 - Erythrocyanin.
124. Proton gradient is broken down due to
- Movement of electrons across the membrane to stroma
 - Movement of electrons across the membrane to lumen
 - Movement of proton across the membrane to lumen
 - Movement of proton across the membrane to stroma
125. By which plant pigment maximum absorption of radiation takes place in the blue and red regions of absorption spectrum?
- Chlorophyll-a
 - Chlorophyll-b
 - Xanthophyll
 - Carotenoid
126. I. H_2S , not H_2O is involved in photosynthesis of sulphur bacteria
 II. ATP is produced during light reaction via chemiosmosis
 III. Absence of light leads to the stoppage of photosynthesis
 IV. Calvin cycle occurs in grana of chloroplast
 Select the correct option
- II, III and IV
 - I, III and IV
 - I, II and IV
 - I, II and III
127. Light Harvesting Complex (LHC) consists of
- One molecule of chlorophyll-a
 - Very few molecules of chlorophyll-a
 - Hundreds of pigment molecules bound to proteins
 - Chlorophyll-a + chlorophyll-c + protein + DNA

128. Which of the following represents the correct molecular formula of chlorophyll-b?
- $C_{55}H_{72}O_6N_4Mg$
 - $C_{55}H_{72}O_5N_4Mg$
 - $C_{55}H_{72}O_4N_4Mg$
 - $C_{55}H_{70}O_6N_4Mg$
129. In C_4 -plants, the bundle sheath cells
- Have thin walls to facilitate gaseous exchange
 - Have large intercellular spaces
 - Are rich in PEP carboxylase
 - Have a high number of chloroplasts
130. Absorption spectrum of chlorophyll- α and the action spectrum of photosynthesis is similar because chlorophyll- α
- Absorbs the maximum light
 - Absorbs the minimum light
 - Absorbs blue and red light
 - Is found most abundantly
131. Sometimes RBCs form a pile or stack known as
- Lacuna
 - Islet
 - Rouleaux
 - Canaliculi.
132. The coagulation of blood occurs due to
- Destruction of erythrocytes
 - Destruction of leucocytes
 - Change of fibrinogen into fibrin
 - Formation of serum.
133. Formation of blood corpuscles is known as
- Haemolysis
 - Rouleaux
 - Haemopoiesis
 - Phagocytosis.
134. Squeezing of leucocytes out from the capillary walls to fight foreign agents is known as
- Diapedesis
 - Diapause
 - Diaphysis
 - Diakinesis.
135. To store blood some anticoagulant is added. It can be
- NaCl
 - Sodium oxalate
 - KCl
 - Thromboplastin.
136. Leucopaenia is the condition in which
- The total number of leucocytes decreases below 5000/cu mm.
 - The total number of leucocytes increases above 6000/cu mm.
 - Abnormal increase in number of WBC
 - Formation of thrombocytes stops.
137. The amount of water present in blood plasma is
- 60%
 - 70%
 - 80%
 - 90%
138. All veins have deoxygenated blood except
- Renal artery
 - Hepatic vein
 - Hepatic portal vein
 - Pulmonary veins.
139. Systemic circulation of oxygenated blood starts from
- Right atrium
 - Left atrium
 - Left ventricle
 - Right ventricle.
140. Murmur occurs due to defect in
- Heart valves
 - SA node
 - AV node
 - Purkinje fibres.
141. Kidney-shaped nucleus occurs in
- Neutrophil
 - Eosinophil
 - Monocyte
 - Lymphocyte.
142. Papillary muscles occur in
- Ventricles
 - Auricles
 - Aortic valve
 - Pulmonary valves.
143. Hardening of arteries due to deposition of cholesterol is
- Atherosclerosis
 - Thrombosis
 - Stenosis
 - Rhinitis.

Date :-17/08/2025

144. Anaemia is due to deficiency of
 (1) Leucocytes
 (2) Blood
 (3) Haemoglobin
 (4) Thrombocytes.
145. Bundle of His is a network of
 (1) Muscle fibres distributed throughout heart walls
 (2) Muscle fibres found only in ventricle wall
 (3) Nerve fibres distributed in ventricles
 (4) Nerve fibres found throughout the heart.
146. Number of leucocytes present in one mm³ of blood is
 (1) 6000 – 8000
 (2) 2000 – 3000
 (3) 8000 – 13000
 (4) 1 million to 1.5 million.
147. Which one prevents blood clotting in blood vessels?
 (1) Heparin (2) Fibrinogen
 (3) Albumin (4) Globulins.
148. Vitamin K is required for
 (1) Blood clotting
 (2) Respiration
 (3) Carbohydrate metabolism
 (4) Calcium phosphorus metabolism.
149. Largest number of white blood corpuscles are
 (1) Eosinophils
 (2) Basophils
 (3) Neutrophils
 (4) Monocytes.
150. The primary CO₂ acceptor in C₃ plants is
 (1) RuBP (2) PGA
 (3) PEP (4) OAA
151. Identify the incorrect match:
 (1) Regeneration of PEP – Mesophyll cells
 (2) RuBP oxygenase activity – Chloroplast
 (3) Photorespiratory loss of CO₂ – Cytoplasm
 (4) Decarboxylation in C₄ pathway – Bundle sheath cell
152. Within the chloroplast there is a membranous system consisting of:
 (1) Grana
 (2) Stroma
 (3) Stroma lamellae
 (4) Both (1) and (3)
153. I. Photorespiration is favoured by high CO₂, low O₂, rise in temperature, high light intensity.
 II. Bundle sheath cells of C₄ plants have agranal chloroplasts, have RuBisCO enzyme and no PEPcase.
 III. Mesophyll cells of C₄ plants have no RuBisCO but have PEPcase.
 IV. Maize and *Sorghum* are C₃ plants.
 Select the correct option for the above statements:
 (1) All are correct
 (2) I and IV are correct
 (3) II and III are correct
 (4) Only IV is correct
154. Which of the following statement is incorrect?
 (1) Site of photosynthesis is chloroplast in plants.
 (2) In Z-scheme, movement of electrons is uphill in terms of redox potential scale.
 (3) In Z-scheme of photosynthesis, the electrons flow from H₂O to NADP⁺
 (4) ATP synthesis is linked to development of a proton gradient across a membrane
155. Which of the following statement is incorrect?
 (1) Reaction centre of photosystem I is P700.
 (2) Photosystem-II receives electrons from photolytic dissociation of water.
 (3) Formation of NADPH is associated with photosystem-II.
 (4) Photosystem-II receives electrons from photosystem-I
156. Which is the first stable product formed during CO₂ fixation in plants like *Sorghum* and maize?
 (1) Oxaloacetic acid
 (2) Pyruvic acid
 (3) Phenyl pyruvic acid
 (4) Phosphoglyceric acid
157. Abnormal rise in erythrocyte count is
 (1) Leucopenia
 (2) Polycythemia
 (3) Anaemia
 (4) Pneumonia.

158. Find out the incorrect statement from the following:
- (1) Dark reaction depends on the product formed by light reaction.
 - (2) In stroma, enzymatic reactions incorporate CO_2 into the plant leading to the synthesis of sugar.
 - (3) Purple and green sulphur bacteria use H_2S as hydrogen donor.
 - (4) There is no division of labour in chloroplast.
159. Identify the incorrect statement regarding xanthophylls and carotenoids pigments in photosynthesis
- (1) They act as accessory pigments
 - (2) Their peak absorption is in blue and red regions of the visible radiations
 - (3) They enable a wider range of wavelength of incoming light to be utilized for photosynthesis
 - (4) They protect chlorophyll-a from photooxidation
160. Identify the statement that is true for C_4 plants, but not for C_3 plants
- Since 1999
- (1) Initial carbon dioxide fixation takes place in bundle sheath cells in the form of 3-carbon compound
 - (2) Initial carbon dioxide fixation takes place in mesophyll cells in the form of 3-carbon compound
 - (3) Initial carbon dioxide fixation takes place in mesophyll cells in the form of 4-carbon compound
 - (4) Initial carbon dioxide fixation takes place in bundle sheath cells in the form of 4-carbon compound
161. The ratio of RBC to WBC in humans is
- (1) 6 : 1
 - (2) 60 : 1
 - (3) 600 : 1
 - (4) 6000 : 1.
162. Which is phagocytic structure of blood ?
- (1) Eosinophil
 - (2) Basophil
 - (3) Platelet
 - (4) Monocyte.
163. Which one produces antibodies ?
- (1) Macrophages
 - (2) Lymphocytes
 - (3) Mast cells
 - (4) Fibroblasts.
164. Origin of heart beat and its conduction is represented by
- (1) A node $\rightarrow$ Bundle of His $\rightarrow$ SA node $\rightarrow$ Purkinje fibres
 - (2) SA node $\rightarrow$ Purkinje fibres $\rightarrow$ AV node $\rightarrow$ Bundle fibres
 - (3) Purkinje fibres $\rightarrow$ AV node $\rightarrow$ SA node $\rightarrow$ Bundle of His
 - (4) SA node $\rightarrow$ AV node $\rightarrow$ Bundle of His $\rightarrow$ Purkinje fibres.
165. Heart beat is accelerated by
- (1) Sympathetic nerves and acetylcholine
 - (2) Vagus nerves and adrenaline
 - (3) Cranial nerves and acetylcholine
 - (4) Sympathetic nerves and epinephrine.
166. Conversion of fibrinogen to fibrin is catalysed by
- (1) Prothrombin
 - (2) Thromboplastin
 - (3) Thrombin
 - (4) All the above.
167. Vitamin required for the formation of prothrombin is
- (1) K
 - (2) B_{12}
 - (3) E
 - (4) C.
168. Thrombocytes have a life of
- (1) 3–4 weeks
 - (2) 4–5 weeks
 - (3) 5–9 days
 - (4) 120 days
169. The one devoid of muscular coat is
- (1) Vein
 - (2) Artery
 - (3) Arteriole
 - (4) Capillary.
170. Glucose is carried from digestive tract to liver by
- (1) Hepatic artery
 - (2) Hepatic portal vein
 - (3) Pulmonary vein
 - (4) Renal vein